Formal derivation of the EM algorithm for a two-component GMM

Model and notation

We use the same notation as in the lecture slides.

For a two-component Gaussian mixture model,

$$\theta = (\pi_1, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2),$$

with

$$\pi_2 = 1 - \pi_1.$$

The latent variable is

$$Z_i \in \{1, 2\}.$$

Conditional on the component,

$$X_i \mid Z_i = k \sim N(\mu_k, \sigma_k^2), \qquad k = 1, 2.$$

We write

$$f(x_i; \mu_k, \sigma_k^2)$$

for the Gaussian density with mean μ_k and variance σ_k^2.

Complete likelihood

Following the notation in the slides,

$$P(X, Z \mid \theta) = \prod_{i=1}^{n} \prod_{k=1}^{2} [P(Z_i = k \mid \theta) P(X \mid Z_i = k, \theta)]^{I(Z_i = k)}.$$

Since

$$P(Z_i = k \mid \theta) = \pi_k$$

and

$$P(X \mid Z_i = k, \theta) = f(x_i; \mu_k, \sigma_k^2),$$

we have

$$P(X, Z \mid \theta) = \prod_{i=1}^{n} \prod_{k=1}^{2} \pi_k^{I(Z_i = k)} f(x_i; \mu_k, \sigma_k^2)^{I(Z_i = k)}.$$

Therefore, the complete log-likelihood is

$$\log(P(X, Z \mid \theta)) = \sum_{i=1}^{n} \sum_{k=1}^{2} I(Z_i = k) \left[\log(\pi_k) + \log\left(f(x_i; \mu_k, \sigma_k^2)\right)\right].$$

For $K = 2$,

$$\begin{aligned}\log(P(X, Z \mid \theta)) = &\sum_{i=1}^{n} I(Z_i = 1) \left[\log(\pi_1) + \log\left(f(x_i; \mu_1, \sigma_1^2)\right)\right] \\ &+ \sum_{i=1}^{n} I(Z_i = 2) \left[\log(1 - \pi_1) + \log\left(f(x_i; \mu_2, \sigma_2^2)\right)\right].\end{aligned}$$

E-step

Let

$$\theta^0 = (\pi_1^0, \mu_1^0, (\sigma_1^2)^0, \mu_2^0, (\sigma_2^2)^0)$$

denote the current value of the parameters.

The E-step computes the posterior distribution

$$P(Z \mid X, \theta^0)$$

and then the expectation of the complete log-likelihood with respect to this posterior:

$$Q(\theta, \theta^0) = E_{Z|X,\theta^0} \left[\log(P(X, Z \mid \theta))\right].$$

Substituting the complete log-likelihood,

$$Q(\theta, \theta^0) = E_{Z|X,\theta^0} \left[\sum_{i=1}^{n} \sum_{k=1}^{2} I(Z_i = k) \Big\{ \log(\pi_k) + \log(f(x_i; \mu_k, \sigma_k^2)) \Big\} \right].$$

By linearity of expectation,

$$Q(\theta, \theta^0) = \sum_{i=1}^{n} \sum_{k=1}^{2} E_{Z|X,\theta^0} \left[I(Z_i = k)\right] \Big\{ \log(\pi_k) + \log(f(x_i; \mu_k, \sigma_k^2)) \Big\}.$$

Now,

$$E_{Z|X,\theta^0} \left[I(Z_i = k)\right] = P(Z_i = k \mid X, \theta^0).$$

Therefore,

$$\boxed{Q(\theta, \theta^0) = \sum_{i=1}^{n} \sum_{k=1}^{2} P(Z_i = k \mid X, \theta^0) \left[\log(\pi_k) + \log(f(x_i; \mu_k, \sigma_k^2))\right]}.$$

For $K = 2$,

$$Q(\theta,\theta^0) = \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)\left[\log(\pi_1) + \log(f(x_i;\mu_1,\sigma_1^2))\right] + \sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)\left[\log(1-\pi_1) + \log(f(x_i;\mu_2,\sigma_2^2))\right].$$

The probabilities used in the E-step are obtained using Bayes' rule.

For component 1,

$$\boxed{P(Z_i = 1 \mid X, \theta^0) = \frac{\pi_1^0 f(x_i;\mu_1^0,(\sigma_1^2)^0)}{\pi_1^0 f(x_i;\mu_1^0,(\sigma_1^2)^0) + (1-\pi_1^0) f(x_i;\mu_2^0,(\sigma_2^2)^0)}}.$$

For component 2,

$$\boxed{P(Z_i = 2 \mid X, \theta^0) = \frac{(1-\pi_1^0) f(x_i;\mu_2^0,(\sigma_2^2)^0)}{\pi_1^0 f(x_i;\mu_1^0,(\sigma_1^2)^0) + (1-\pi_1^0) f(x_i;\mu_2^0,(\sigma_2^2)^0)}}.$$

During the M-step, these probabilities are treated as fixed.

M-step

The M-step obtains new parameter estimates

$$\hat{\theta} = \arg\max_{\theta} Q(\theta,\theta^0).$$

We now maximise $Q(\theta,\theta^0)$ with respect to each parameter.

Derivation of $\widehat{\mu}_1$

The terms involving μ_1 are

$$\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)\log(f(x_i;\mu_1,\sigma_1^2)).$$

For a Gaussian density,

$$\log(f(x_i;\mu_1,\sigma_1^2)) = -\frac{1}{2}\log(2\pi\sigma_1^2) - \frac{(x_i-\mu_1)^2}{2\sigma_1^2}.$$

Therefore,

$$\frac{\partial Q(\theta,\theta^0)}{\partial \mu_1} = \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)\frac{x_i-\mu_1}{\sigma_1^2}.$$

Set this equal to zero:

$$\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)(x_i - \mu_1) = 0.$$

Hence,

$$\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) x_i = \mu_1 \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0).$$

Therefore,

$$\boxed{\widehat{\mu}_1 = \frac{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) x_i}{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}}.$$

Similarly,

$$\boxed{\widehat{\mu}_2 = \frac{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0) x_i}{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)}}.$$

Derivation of $\widehat{\sigma}_1^2$

The terms involving σ_1^2 are

$$-\frac{1}{2} \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) \left[\log(\sigma_1^2) + \frac{(x_i - \mu_1)^2}{\sigma_1^2} \right].$$

Differentiate with respect to σ_1^2:

$$\frac{\partial Q(\theta, \theta^0)}{\partial \sigma_1^2} = -\frac{1}{2} \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) \left[\frac{1}{\sigma_1^2} - \frac{(x_i - \mu_1)^2}{(\sigma_1^2)^2} \right].$$

Set this equal to zero:

$$\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) \left[\frac{1}{\sigma_1^2} - \frac{(x_i - \mu_1)^2}{(\sigma_1^2)^2} \right] = 0.$$

Multiplying by $(\sigma_1^2)^2$ gives

$$\sigma_1^2 \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) = \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)(x_i - \mu_1)^2.$$

Therefore,

$$\boxed{\widehat{\sigma}_1^2 = \frac{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)(x_i - \widehat{\mu}_1)^2}{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}}.$$

Similarly,

$$\boxed{\widehat{\sigma}_2^2 = \frac{\sum_{i=1}^n P(Z_i = 2 \mid X, \theta^0)(x_i - \widehat{\mu}_2)^2}{\sum_{i=1}^n P(Z_i = 2 \mid X, \theta^0)}}.$$

Derivation of $\widehat{\pi}_1$

Because $K = 2$,

$$\pi_2 = 1 - \pi_1.$$

The part of $Q(\theta, \theta^0)$ involving π_1 is

$$\sum_{i=1}^n \left[P(Z_i = 1 \mid X, \theta^0) \log(\pi_1) + P(Z_i = 2 \mid X, \theta^0) \log(1 - \pi_1) \right].$$

Differentiate with respect to π_1:

$$\frac{\partial Q(\theta, \theta^0)}{\partial \pi_1} = \sum_{i=1}^n \left[\frac{P(Z_i = 1 \mid X, \theta^0)}{\pi_1} - \frac{P(Z_i = 2 \mid X, \theta^0)}{1 - \pi_1} \right].$$

Set this equal to zero:

$$\frac{\sum_{i=1}^n P(Z_i = 1 \mid X, \theta^0)}{\pi_1} = \frac{\sum_{i=1}^n P(Z_i = 2 \mid X, \theta^0)}{1 - \pi_1}.$$

Since

$$P(Z_i = 1 \mid X, \theta^0) + P(Z_i = 2 \mid X, \theta^0) = 1,$$

we obtain

$$\boxed{\widehat{\pi}_1 = \frac{1}{n} \sum_{i=1}^n P(Z_i = 1 \mid X, \theta^0)}.$$

Since $\pi_2 = 1 - \pi_1$,

$$\boxed{\widehat{\pi}_2 = \frac{1}{n} \sum_{i=1}^n P(Z_i = 2 \mid X, \theta^0)}.$$

Final EM updates for $K = 2$

Starting from the current parameter values θ^0:

E-step

Compute

$$P(Z_i = 1 \mid X, \theta^0) = \frac{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0)}{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0) + (1 - \pi_1^0) f(x_i; \mu_2^0, (\sigma_2^2)^0)},$$

and

$$P(Z_i = 2 \mid X, \theta^0) = 1 - P(Z_i = 1 \mid X, \theta^0).$$

M-step

With these probabilities fixed,

$$\boxed{\widehat{\pi}_1 = \frac{1}{n} \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}$$

$$\boxed{\widehat{\mu}_1 = \frac{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) x_i}{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}}$$

$$\boxed{\widehat{\mu}_2 = \frac{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0) x_i}{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)}}$$

$$\boxed{\widehat{\sigma}_1^2 = \frac{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)(x_i - \widehat{\mu}_1)^2}{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}}$$

and

$$\boxed{\widehat{\sigma}_2^2 = \frac{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)(x_i - \widehat{\mu}_2)^2}{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)}}.$$

The new estimate is therefore

$$\hat{\theta} = (\widehat{\pi}_1, \widehat{\mu}_1, \widehat{\sigma}_1^2, \widehat{\mu}_2, \widehat{\sigma}_2^2).$$

We then replace the current parameter values by the new estimates,

$$\theta^0 \leftarrow \hat{\theta},$$

and repeat the E-step and M-step until the incomplete log-likelihood changes by less than a small value ϵ.